#9 · 4.0% of marks

Ch 2 — Risk Management

Short chapter — learn every list. SM TYK theory Qs: **significance of VaR** (3.2) · **financial risk from different perspectives** (2). Numerical: VaR for a single share / two-share portfolio.

1 Types of risk faced by an organisation (SM 1)

Risk	SM meaning & example
Strategic	*Company's strategy becomes less effective and it struggles to reach its goal — technology change, new competitor, shift in demand, input-cost rise.* ***Kodak*** *(built a digital camera in 1975, saw it as a threat, left behind) ·* ***Nokia*** *(late to touch screens → Samsung led) ·* ***Xerox*** *(positive — moved into laser printing)*
Compliance	*Complying with Companies Act 2013, SEBI etc.; non-compliance → fine / imprisonment. Real problem when entering a* ***new business line or geography*** *(cement company entering sugar in another state)*
Operational	***Internal****; failure to cope with day-to-day problems; relates to* ***people and process*** *(paying ₹1,00,000 instead of ₹10,000 → second checker or system flag)*
Financial	*Unexpected changes in prices, exchange rates, credit rating, interest rates; political risk included as it leads to country risk*

2 Categories of financial risk (SM 1.4)

- → ***Counter-party****: non-honouring of obligations (goods not delivered after payment, loan / interest not repaid) — includes* ***credit risk***
- → ***Political****: faced by overseas investors — (i) confiscation / destruction of property (ii) rationing of remittance (iii) restriction on currency conversion (iv) restriction on borrowings (v) invalidation of patents (vi) price control*
- → ***Interest rate****: rate changes alter assets & liabilities; most important for* ***banks*** *(earn the spread). Inherent in both fixed and floating rates — floating borrower suffers when rates rise; fixed borrower pays comparatively more when floating rates fall*
- → ***Currency****: cash flows change with exchange rates, adversely or favourably — ₹ depreciation helps an exporter (****Infosys****), hurts an importer (****IOC****)*
- → ***Liquidity****: inability to meet liabilities when due — inadequate cash or mismatch in timing; common in banks (maturity mismatch)*

3 Evaluation of financial risk — 3 views (SM 2) — TYK Q2

View	SM
Stakeholders	*Equity holders see* ***financial gearing*** *as risk (last in winding up); lenders see high gearing as risk of default on interest & principal*
Company	*Excessive borrowing, or lending to someone who defaults, can force* ***liquidation***
Government	*Failure of a bank (****Lehman Brothers****) or downgrade of a financial institution spreads distrust; includes* ***wilful defaulters****; extends to* ***sovereign debt crisis***

TYK THEORY Q1 + NUMERICAL

④ Value at Risk (SM 3)

*Measure of investment risk: under normal market conditions over a set period (say one day), **how much an investment might lose** — portfolio, capital investment or forex. Answers: (i) what is the **worst case**? (ii) what will be the **loss**? First applied **1922 at NYSE**; entered finance in the **1990s**; now the world's most widely used measure of financial risk.*

Features (SM 3.1)

*(i) **Components**: time period · confidence level (generally **95% / 99%**) · loss in % or amount (ii) **statistical** tool based on standard deviation (iii) any **time horizon** — day, week, month (iv) **probability** of maximum loss, assuming normal distribution (v) **risk control** by setting maximum-loss limits (vi) **Z score** = number of σ from the mean; Z × σ = VaR*

Application / significance (SM 3.2) — TYK Q1

*(a) measure **maximum possible loss** on a portfolio or trading position (b) **benchmark** for performance measurement of operations / trading (c) fix **limits for front-office** dealers in treasury (d) help management decide **trading strategies** (e) tool for **ALM**, especially in banks*

VaR = Z × σ (in ₹) × √t · Z = **2.33** at 99%, **1.65** at 95% (one-tailed) · $\sigma_{t\ days} = \sigma_{1\ day} \times \sqrt{t}$

SM working	Answer
SM 3.3: ₹2 cr in X Ltd, daily σ 2%, 252 trading days, 99%	σ = ₹4 lakh → 1-day VaR = 4 × 2.33 = **₹9.32 lakh**; 10-day = √10 × 9.32 = **₹29.47 lakh**
TYK 1: ₹200 lakh each in XYZ & ABC, daily σ 1% each, r = 0.3, 10-day 99%	Daily σ each ₹2 lakh; variance = $2^2 + 2^2 + 2 \times 0.3 \times 2 \times 2 = 10.4$ → σ = **₹3.22 lakh** (or 0.80623% of ₹400 lakh); 10-day σ = 3.22 × √10 = **₹10.18 lakh**; VaR = 2.33 × 10.18 = **₹23.72 lakh**

⚠ TRAP

*Scale σ by **√t**, never by t. Use portfolio σ with the correlation term — two 1% shares with r = 0.3 give 0.806%, not 1%. 252 days is only needed to annualise.*

⑤ Identifying & managing financial risk (SM 4)

Counter-party risk (4.1)

***Hints (5)**: (a) failure to obtain resources to complete the project (b) regulatory restrictions (c) hostile action of a foreign government (d) let down by a third party (e) insolvency. **Management (6)**: (1) **due diligence** before dealing (2) don't over-commit to one entity / group (3) know exposure limits (4) review credit limits & approval procedure regularly (5) rapid action on likely default (6) performance guarantees, insurance, other instruments.*

Political risk (4.2)

Identify (4): insistence on resident investors / labour · restriction on currency conversion · **expropriation** of foreign assets · price fixation

Assess country risk (4): political rankings in business magazines · macro-economic conditions · popularity & stability of government · advice from home-country **embassies**

Mitigate (4): **local sourcing** of materials & labour · **joint ventures** · **local financing** · **prior negotiations**

Interest rate risk (4.3) — identified from (6)

Monetary policy · government action (eg **demonetisation**) · economic growth · release of industrial data · foreign investment · stock-market changes. Management → Ch 12.

Currency risk (4.4) — parameters (6)

(1) **Government action** (Brexit → £ lowest since the 1980s) (2) **nominal interest rate** (IRP) (3) **inflation** (PPP) (4) natural calamities (5) war, coup, rebellion (6) change of government and its attitude to foreign investment. Management → Ch 10.

Mnemonic for currency parameters: **G-I-I-N-W-C** — Government, Interest, Inflation, Natural calamity, War, Change of government.

⑥ Correlation — how Ch 2 actually connects

Ch 2 idea	The actual link (app-written cross-reference; the Ch 2 side is SM)
VaR of a 2-share portfolio	**Ch 6**: $\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2r\sigma_1\sigma_2$ — the same formula; lower r → lower σ_p → lower VaR. That is diversification expressed in rupees.
Interest rate risk	**Ch 12**: identified here, managed there (gap, FRA, caps, swaps); **Ch 5**: price vs reinvestment risk, duration.
Currency risk — IRP, PPP	**Ch 10**: the parity conditions forecast the rate; the hedges (forward, MMH, options) manage it; **Ch 11**: political-risk list = complexity list in international capital budgeting.
Political risk mitigation	**Ch 11**: local financing and JVs reduce blocked-fund and expropriation risk; certainty-equivalent cash flows rather than a higher discount rate.
Operational risk, limits	**Ch 9**: Barings (back office run by the trader), Orange County — lessons = separate front / middle / back office, risk limits, VaR + stress tests.
Financial gearing as risk	**FM / Ch 14**: high gearing raises K_e; LBO targets carry the most gearing risk; **AAA**: SA 570 going-concern indicators (defaults, negative net worth).

② Risk Management — what ICAI asked (Sep 2025, Jan 2026, May 2026)

14 marks *set across the last three attempts (4.0% of marks set). Question labels and marks are ICAI's; the one-line concept descriptions are app-written.*

Attempt	Question	Marks	Concept tested
Sep 2025	Q4(c) [or]	4	Parameters used to identify currency risk (SM Ch 2 section on currency risk) (Theory)
Jan 2026	Q3(c)	4	Political risk: host-government actions that signal it, and techniques to mitigate it (SM Ch 2 section on political risk) (Theory)
May 2026	MCQ CS-IV 10-12	6	Value at Risk of a two-stock portfolio (annual VaR at 99%), diversification benefit at 90%, effect of change in correlation (MCQ)

What follows, exactly as ICAI printed it in the AFM study material: (1) Test Your Knowledge theory questions, with the SM paragraph ICAI's answer points to; (2) every SM illustration with ICAI's solution, grouped by SM section; (3) Test Your Knowledge practical questions with ICAI's solutions.

Ⓣ Theory questions (SM Test Your Knowledge) with ICAI answers

Theory Q1

1. Explain the significance of VAR.

ICAI answer: "Please refer paragraph 3.2" – SM paragraph 3.2 reproduced

3.2 Application of VAR

VAR can be applied

(a) to measure the maximum possible loss on any portfolio or a trading position.

(b) as a benchmark for performance measurement of any operation or trading.

(c) to fix limits for individuals dealing in front office of a treasury department.

(d) to enable the management to decide the trading strategies.

(e) as a tool for Asset and Liability Management especially in banks.

ICAI answer: "Please refer paragraph 3.2" – SM paragraph 3.2 reproduced

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Theory Q2

2. The Financial Risk can be viewed from different perspective. Explain.

ICAI answer: "Please refer paragraph 2" – SM paragraph 2 reproduced

2. EVALUATION OF FINANCIAL RISK

The financial risk can be evaluated from different point of views as follows:

(a) From stakeholder's point of view: Major stakeholders of a business are equity shareholders and they view financial gearing i.e. ratio of debt in capital structure of company as risk since in event of winding up of a company they will be least prioritized.

Even for a lender, existing gearing is also a risk since company having high gearing faces more risk in default of payment of interest and principal repayment.

(b) From Company's point of view: From company's point of view if a company borrows excessively or lend to someone who defaults, then it can be forced to go into liquidation.

(c) From Government's point of view: From Government's point of view, the financial risk can be viewed as failure of any bank or (like Lehman Brothers) down grading of any financial institution leading to spread of distrust among society at large. Even this risk also includes wilful defaulters. This can also be extended to sovereign debt crisis.

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Ⓟ Practical questions (SM Test Your Knowledge) with ICAI solutions

Practical Q1

1. Consider a portfolio consisting of a ₹ 200,00,000 investment in share XYZ and a ₹ 200,00,000 investment in share ABC. The daily standard deviation of both shares is 1% and that the coefficient of correlation between them is 0.3. You are required to determine the 10-day 99% value at risk for the portfolio?

Solution — ICAI

1. The standard deviation of the daily change in the investment in each asset is ₹ 2,00,000 i.e. 2 lakhs. The variance of the portfolio's daily change is

$$V = 2^2 + 2^2 + 2 \times 0.3 \times 2 \times 2 = 10.4$$

$$\sigma \text{ (Standard Deviation)} = \sqrt{10.4} = ₹ 3.22 \text{ lakhs}$$

Alternatively, it can also be computed as follows:

$$= (1)^2(0.50)^2 + (1)^2(0.50)^2 + 2(1)(1)(0.3)(0.50)(0.50)$$

$$= 0.25 + 0.25 + 0.15 = 0.65\%$$

$$\sigma \text{ (Standard Deviation)} = \sqrt{0.65} = 0.80623\%$$

σ (Standard Deviation) in Amount = ₹ 400 lakhs x 0.80623% = ₹ 3.22 lakhs

Accordingly, the standard deviation of the 10-day change is

$$₹ 3.22 \text{ lakhs} \times \sqrt{10} = ₹ 10.18 \text{ lakh}$$

From the Normal Table we see that z score for 1% is 2.33. This means that 1% of a normal distribution lies more than 2.33 standard deviations below the mean. The 10-day 99 percent value at risk is therefore

$$2.33 \times ₹ 10.18 \text{ lakh} = ₹ 23.72 \text{ lakh}$$

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